

CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

DPP- 02 Discussion Notes



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[NAT]

$${}^nC_2 \Rightarrow \frac{{}^nC_2}{{}^nC_2} = \frac{n(n-1)}{2} = \frac{7 \cdot 6}{2} = 21$$



#Q. Let G is simple graph with 7 vertices and 11 edges. Then find number of edges in complement of G

$$\begin{aligned} E(G) + E(\bar{G}) &= {}^nC_2 = 7C_2 \Rightarrow \frac{7 \cdot 6}{2} = 21 \\ 11 + x &= 21 \\ x &= 21 - 11 = \underline{\underline{10}} \end{aligned}$$

#Q. In a self-complementary graph G of size 18, then find the number of vertices in the graph G ?

SC graph - A graph which is isomorphic to its complement.

$$E(G) + E(\bar{G}) = nC_2 = \frac{n(n-1)}{2}$$

$$\downarrow \quad \downarrow$$

$$x + x \Rightarrow 2x = \frac{n(n-1)}{2}$$

$$2 \cdot 2 \cdot 18 = n^2 - n$$

$$\Rightarrow n^2 - n - 72 = 0 \quad 8 \times 9 = 72$$

$$n^2 - 8n + 9n - 72 = 0$$

$$n(n-8) + 9(n-8) = 0$$

$$n^2 - 8n + 9n - 72 = 0$$

$$n(n-8) + 9(n-8) = 0$$

$$8(n-9) = 0 \Rightarrow$$

$$n = 9$$

$$n = -8$$

#Q. If G is a simple graph with degree sequence $\{5, 2, 2, 2, 2, 1\}$ then what is the number of edges in the complement $\bar{G}$? Also identify the degree sequence for the complement of graph G .

$$\sum d(v) = 2 \cdot E$$

$$5 + 2 + 2 + 2 + 2 + 1 = 2 \cdot E$$

$$14 = 2 \cdot E \Rightarrow E = 7$$

$$E(G) + E(\bar{G}) = \frac{n(n-1)}{2}$$

$$7 + \frac{n(n-1)}{2} = \frac{n(n-1)}{2}$$

$$7 + \frac{n(n-1)}{2} = \frac{n(n-1)}{2} \Rightarrow n = 8$$

degree in $\bar{G} = n - 1 - d$

$\Rightarrow 8 - 1 - 5 = 2$
 $8 - 1 - 2 = 5$
 $8 - 1 - 2 = 5$
 $8 - 1 - 2 = 5$
 $8 - 1 - 2 = 5$
 $8 - 1 - 1 = 6$

$\Rightarrow 2, 5, 5, 5, 5, 6$

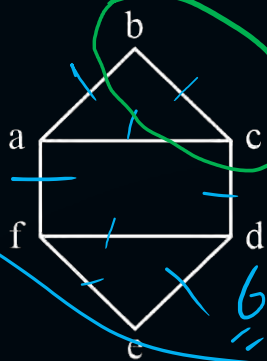
$\{2, 5, 5, 5, 5, 6\}$

[MSQ]



#Q. Which of the following options is/are correct for isomorphic graphs?

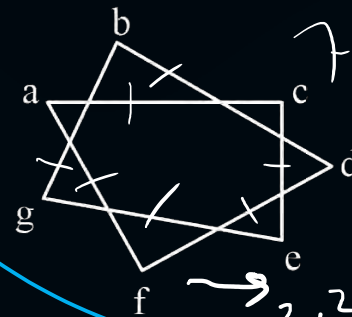
G_1 :



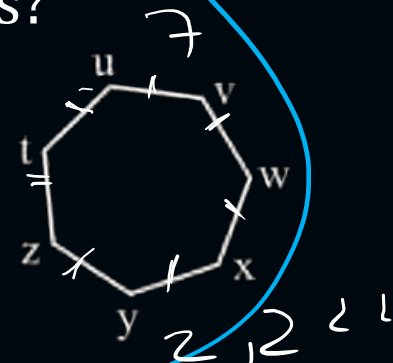
G_2 :



G_3 :



G_4 :



A

~~G_1 and G_2 are isomorphic graph.~~

B

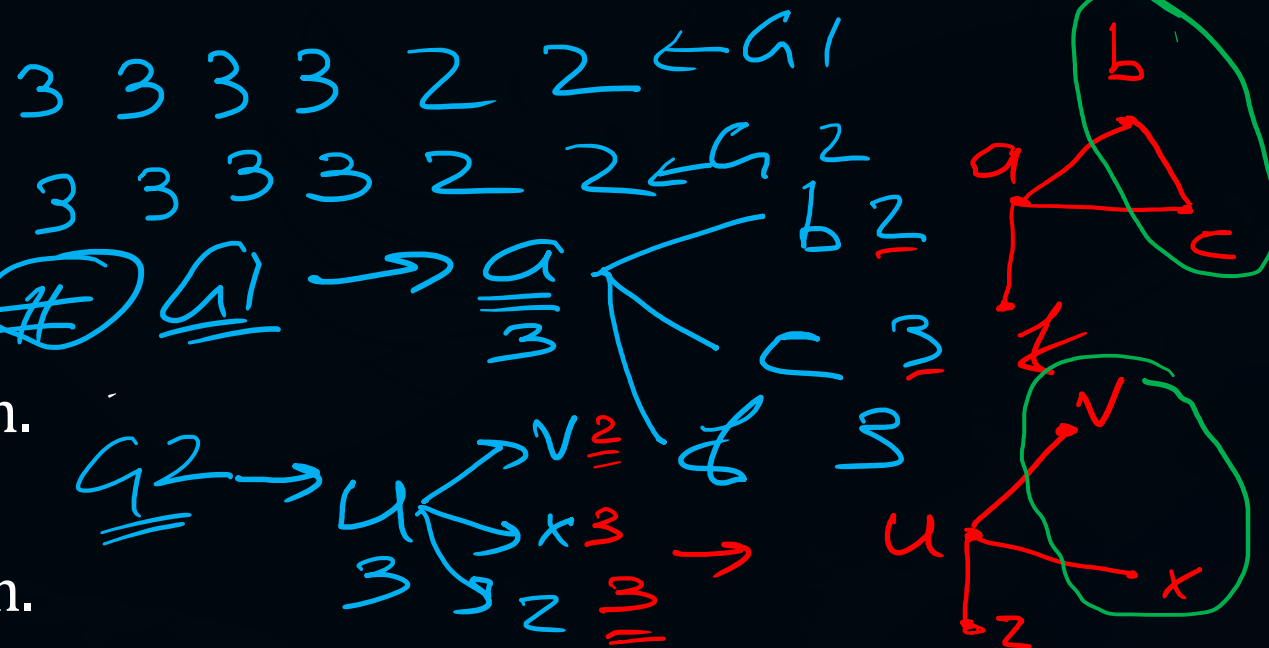
G_3 and G_4 are isomorphic graph.

C

G_1 and G_2 are not isomorphic graph.

D

G_3 and G_4 are not isomorphic graph.



[NAT]

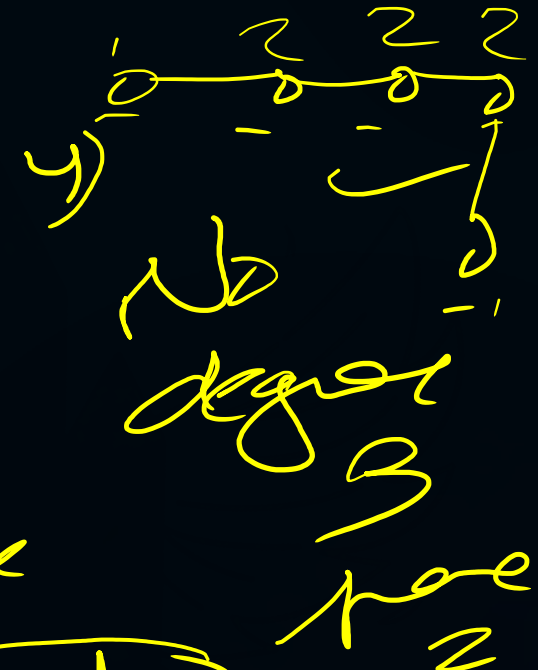


#Q. How many simple non-isomorphic trees are possible with 5-vertices?

1. The ~~edges~~ vertex ~~can~~ be connected.
2. There should not be any cycle.



3) Degree 2



3

Nobel prize
#Tutoring answered

#Q. Let G be a connected planar graph with 20 vertices each of degree 3 then the number of faces in the planar embedding of the graph is _____.

$$\sum d(V) = 2 \cdot E$$

$$20 \cdot 3 = 2E \Rightarrow E = 30$$

$$V - E + F = 2$$

$$20 - 30 + F = 2$$

$$-10 + F = 2 \Rightarrow F = 2 + 10 = \underline{\underline{12}}$$

#Q. Let G be a connected planar graph with 35 regions each of degree 6. Then the number of vertices in graph G is _____.

$$\begin{array}{r} 105 \\ - 35 \\ \hline 70 \end{array}$$

$$\sum d(V) = 2E$$

$$\sum d(R) = 2 \cdot E$$

$$35 \times 6 = 2 \cdot E$$

$$E = 105$$

$$V - E + F = 2$$

$$V - 105 + 35 = 2$$

$$V - 105 + 35 = 2$$

$$V - 70 = 2$$

$$V = 72$$

[NAT]



$$V=12, E=30$$

#Q. Let G be a connected planar graph with 12 vertices and 30 edges, and degree of each region is k . Then the value of k is ____.

$$\sum d(V) = 2 \cdot E \leftarrow$$
$$12 \times 2 = 2 \cdot 30$$

#

$$\sum d(R) = 2 \cdot E$$

$$x \cdot k = 2 \cdot 30$$

$$x \cdot k = 60$$

$$\Rightarrow 20 \cdot k = 60$$
$$k = 3$$

$$\Rightarrow V - E + R = 2$$

$$12 - 30 + R = 2$$

$$R = 2 - 12 + 30$$

$$R = -10 + 30$$
$$R = 20 = x$$

#Q. Minimum number of vertices necessary in a simple connected planar graph with 11 edges is ____.

$$3n \leq 2e$$

$$V - e + f = 2$$

$$V - e + 1 = 2$$

$$e \leq 3V - 6$$

$$11 \leq 3V - 6$$

$$3V \geq 17$$

$$V \geq \frac{17}{3} \approx 5.67$$

$$V \geq 6$$

$$V = 6$$

#Q. Maximum number of regions possible in a simple connected planar graph with 10 vertices is ____.

$$3n \leq 2e$$

$$3n \leq 2 \cdot 24$$

$$n \leq 16$$

$$n = 16$$

$$e \leq 3V - 6$$

$$e \leq 3 \cdot 10 - 6$$

$$e \leq 24$$

[MCQ]



$$\sum d(v) = 2 \cdot E$$
$$5 \cdot 10 = 2 \cdot E$$
$$E = 25$$

#Q. Consider a 5-regular connected planar graph with 10 vertices.
How many bounded faces are there in the planar embedding of the graph?

A 15

C 17

B 16

D 18

$$V - E + F = 2$$
$$10 - 25 + F = 2$$
$$F = 2 - 10 + 25$$
$$F = 17$$
$$F = 17 - 1$$
$$F = 16$$
$$10 \times \frac{5}{2} = 25$$

Unbounded region



All the best

THANK - YOU

